PROMPT EVALUATION, COMPARISON AND REFINEMENT USING LLM RESPONSES

Aim

To design multiple prompts for a text-generation task, evaluate their outputs using predefined criteria, and refine the best-performing prompt.

Procedure

1. Choose a text-generation task: write a short college notice about saving electricity.

2. Write Prompt A with a broad instruction and Prompt B with audience, length, content, style, and format requirements.

3. Generate an output for each prompt using the same LLM.

4. Evaluate each response for relevance, accuracy, completeness, clarity, and format adherence on a 1–5 scale.

5. Compare scores and identify the stronger prompt.

6. Refine the stronger prompt by adding missing constraints and a final self-check.

7. Generate a new response using the refined prompt and evaluate it again.

Structured Prompt

```
Task: Write a notice about saving electricity.
Audience: College students.
Length: 70-90 words.
Required content:
- A clear heading.
- Three practical actions.
- A closing reminder.
Style: Simple, clear, and suitable for a college notice board.
Accuracy: Do not invent statistics or unsupported claims.
Format: Heading followed by one short paragraph.
Before finalizing, check relevance, accuracy, completeness,
clarity, and format adherence.
```

Program / Demonstration

```
task = "Write a short notice about saving electricity."

output_a = ("Save electricity. Turn off lights and fans when not needed. "
            "Use energy-efficient devices.")
output_b = ("NOTICE: SAVE ELECTRICITY\\nCollege students can save energy by "
            "switching off lights and fans when leaving a room, using daylight "
            "when possible, and reporting faulty switches. Use efficient devices "
            "where available. Let us use electricity responsibly.")

criteria = ["relevance", "accuracy", "completeness", "clarity", "format"]
scores = {"Prompt A": [4, 5, 2, 4, 2], "Prompt B": [5, 5, 5, 5, 5]}
print("Prompt A:", output_a)
print("\\nPrompt B:", output_b)
print("\\nScores out of 25:")
for name, values in scores.items():
    print(name, sum(values), "/ 25")
print("Best prompt: Prompt B")
```

Evaluation Table

Illustrative sample scores (1 = poor, 5 = excellent), included to demonstrate the evaluation method.

Prompt	Relevance	Accuracy	Completeness	Clarity	Format	Total
A: broad	4	5	2	4	2	17/25
B: structured	5	5	5	5	5	25/25

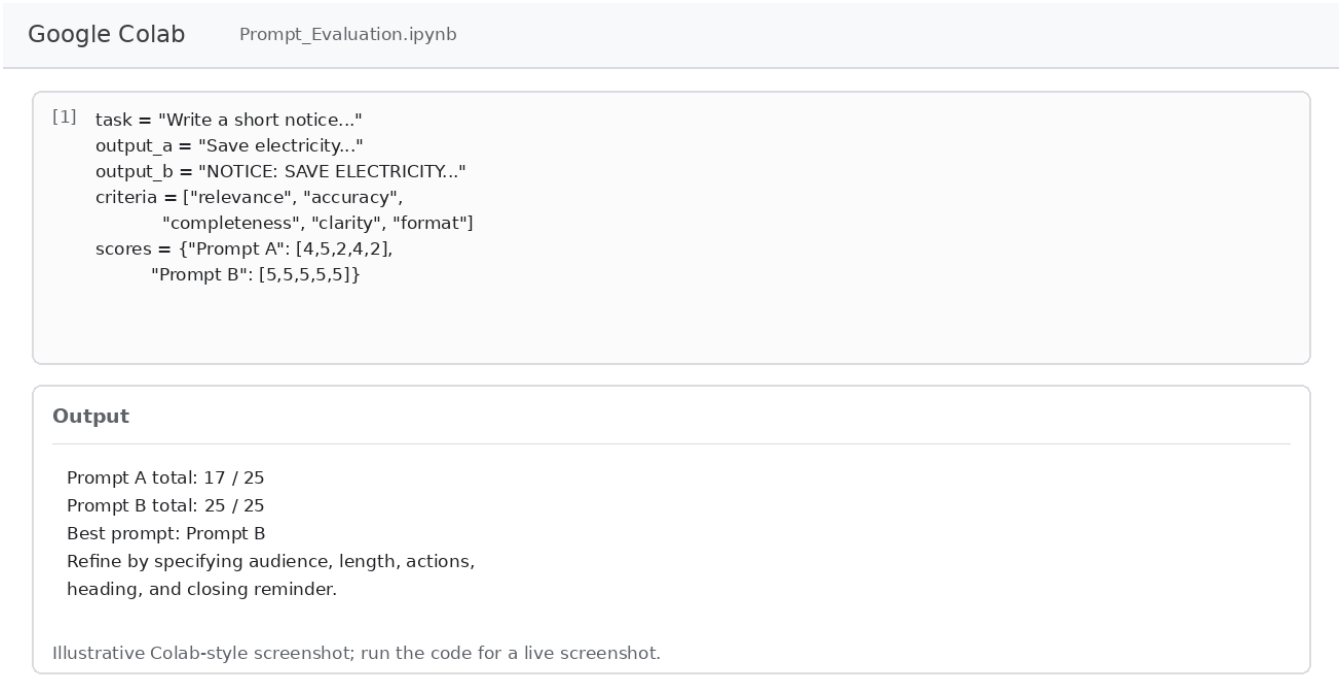
Refined Prompt

```
Task: Write a notice about saving electricity.
Audience: College students.
Length: 70-90 words.
Required content:
- A clear heading.
- Three practical actions.
- A closing reminder.
```

```
Style: Simple, clear, and suitable for a college notice board.
Accuracy: Do not invent statistics or unsupported claims.
Format: Heading followed by one short paragraph.
Before finalizing, check relevance, accuracy, completeness,
clarity, and format adherence.
```

Google Colab Output Screenshot

Illustrative screenshot only, not captured from a live Colab session. Run the code in Colab for an authentic screenshot if required.



Expected Output

```
Prompt A total: 17 / 25
Prompt B total: 25 / 25
Best prompt: Prompt B
Refine by specifying audience, length, actions, heading, and closing reminder.
```

Result

The structured prompt performs better in this illustrative evaluation because it states the audience, length, required content, and format. Measurable constraints can improve clarity and completeness. Actual scores should be assigned after reviewing the LLM's real outputs.